	$= 0.20 \text{ ms cm}^{-1}$
3(a)	The electric potential at a point in an electric field is defined as the work done per unit positive charge in bringing a small test charge from infinity to that point.
3(b)(i)	Using $V = \dfrac{Q}{4\pi\varepsilon_0 r}$, $Q = 4\pi\varepsilon_0 rV$ $Ne = 4\pi\varepsilon_0 rV$ $N = \dfrac{4\pi\varepsilon_0 rV}{e}$ At $r = 2.0\times10^{-10}$ m, $V = 390$ V $N = \dfrac{4\pi(8.85\times10^{-12})(2.0\times10^{-10})(390)}{1.60\times10^{-19}} = 54.22$ At $r = 6.0\times10^{-10}$ m, $V = 130$ V $N = \dfrac{4\pi(8.85\times10^{-12})(6.0\times10^{-10})(130)}{1.60\times10^{-19}} = 54.22$ Hence, the number of protons in the nucleus is 54.
3(b)(ii)	From Fig. 3.1, the radius of the nucleus is assumed to be approximately 1.0×10^{-10} m. The radius of the proton would be smaller. The distance of 2.0×10^{-8} m is about 200 times larger than this radius. So it may be assumed that the proton and the nucleus behave as point charges, such that the electric force between the two particles would not be affected by the charge distribution within each particle.
3(b)(iii)	Where the proton is at distance $r = 2.0\times10^{-8}$ m from the nucleus, the repulsive electric force F_E on the proton results in its acceleration a given by $F_E = m_p a$ $F_E = \dfrac{Qq}{4\pi\varepsilon_0 r^2} = \dfrac{54e^2}{4\pi\varepsilon_0 r^2} = m_p a$ $a = \dfrac{54e^2}{4\pi\varepsilon_0 r^2 m_p} = \dfrac{54(1.60\times10^{-19})^2}{4\pi(8.85\times10^{-12})(2.0\times10^{-8})^2(1.67\times10^{-27})}$ $= 1.861 \times 10^{16}$

No.	Solution
	$\approx 1.9 \times 10^{16}$ m s^{-2}
3(b)(iv)	Loss in electric potential energy ΔU = Increase in kinetic energy ΔE_k Since the proton's electric potential energy at infinity is zero, $\Delta E_k = \Delta U = \dfrac{Qq}{4\pi\varepsilon_0 r} - 0 = \dfrac{54e^2}{4\pi\varepsilon_0 r}$ $= \dfrac{54(1.60 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(2.0 \times 10^{-8})}$ $= 6.215 \times 10^{-19}$ $\approx 6.2 \times 10^{-19}$ J
4(a)(i)	The magnetic force acting on the proton is vertically upwards as it enters the region between the parallel plates. In order to travel through the fields without